

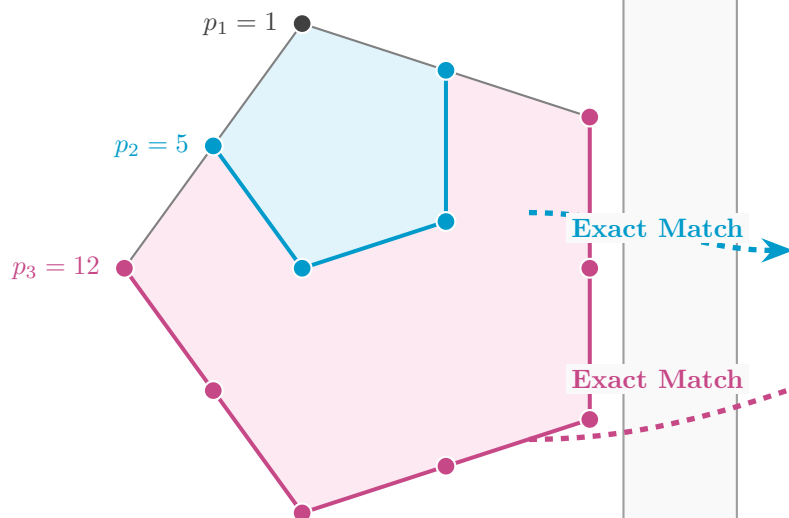
Euler's Pentagonal Number Theorem

Connecting Geometric Pentagons to Combinatorial Partitions

Geometric Origin

Nested pentagons generate the sequence

$$p_k = \frac{k(3k-1)}{2}$$



Combinatorial Exceptions

Franklin's involution pairs even/odd partitions.

It fails **only** when Base and Slope intersect.

- **Base:** Bottom row
- **Slope:** Right diagonal
- **Intersection:** Dot is in both Base and Slope

$n = 5$ dots

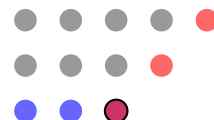
Partition $\lambda = (3, 2)$



Base $\leq$ Slope, but moving the base breaks the slope.
Un-cancelable term: $-x^5$

$n = 12$ dots

Partition $\lambda = (5, 4, 3)$



Base $\leq$ Slope, but moving the base breaks the slope.
Un-cancelable term: $+x^{12}$

The Pentagonal Number Theorem (Euler's Identity)

$$\prod_{n=1}^{\infty} (1 - x^n) = 1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + \dots$$